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# RecoMart: An End-to-End MLOps Pipeline for Product Recommendation

Design and Initial Implementation Report

**Course:** Data Management for Deep Learning
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## Team Members

| Name | Student ID | Contribution |
|---|---|---|
| Team Member 1 | Student ID | Data ingestion, validation, and documentation |
| Team Member 2 | Student ID | Feature engineering, modeling, and evaluation |
| Team Member 3 | Student ID | MLOps architecture, orchestration, and reporting |

# Abstract

RecoMart is an academic recommendation-system project demonstrating how data-management and machine-learning engineering practices combine in a reproducible pipeline. It uses MovieLens 100K to construct customer, product, interaction, purchase, and popularity assets; validates and prepares them; engineers recommendation features; and trains collaborative and content-based recommendation models.

This first report version documents the architecture, processing stages, design rationale, reproducibility principles, and known limitations. It deliberately contains authored static content only. Runtime metrics, screenshots, experiment records, orchestration results, feature-store queries, and dynamically discovered pipeline artifacts are out of scope.

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# 1. Introduction

Recommendation systems help users navigate large catalogues by ranking items according to predicted relevance. Building a useful model is only one part of the problem: supporting data must be collected, validated, transformed, versioned, and presented through a repeatable engineering process. RecoMart addresses this broader lifecycle as a compact, academically inspectable MLOps project.

The repository follows a layered design in which each processing stage has a focused responsibility and publishes explicit artifacts for the next stage. This improves traceability, reduces accidental coupling, and makes failures easier to locate and reason about.

# 2. Project Objectives and Scope

The primary objective is to demonstrate a maintainable recommendation workflow from source-data generation through model evaluation while preserving clear contracts between stages.

- Create realistic recommendation inputs from MovieLens 100K.
- Preserve immutable raw inputs and validate schema and business rules.
- Prepare analysis-ready datasets and reusable recommendation features.
- Train and compare collaborative and content-based approaches.
- Maintain reproducibility through configuration and deterministic processing.
- Document assumptions, decisions, evidence, and limitations clearly.

# 3. Dataset and Problem Definition

RecoMart is based on MovieLens 100K, a benchmark containing user ratings for films. Films represent products and ratings provide the central user-item preference signal. Generated assets supplement the source with customer profiles, product metadata, clickstream-style interactions, purchase-history records, and external-style popularity values.

The learning problem is to rank or score products for a user using past preference signals and product attributes. Evaluation must distinguish predictive performance from catalogue coverage and acknowledge the cold-start difficulty associated with unseen users and products.

# 4. System Architecture

The logical data flow is Incoming to Raw to Validated to Prepared to Features to Models to Reports. Each transition has a dedicated Python package under src, while configuration remains external to processing code. This prevents downstream stages from silently bypassing validation.

## Source generation and ingestion

Source generation converts MovieLens records into RecoMart domain assets. File ingestion handles users, products, clickstream, and purchases, while popularity is collected through a REST endpoint. Raw publication retains source bytes, checksums, and batch metadata.

### Validation and preparation

Validation checks required fields, types, ranges, uniqueness, referential integrity, timestamps, and cross-dataset consistency. Preparation cleans accepted records, reconciles entities, joins optional popularity enrichment, and creates chronological splits.

### Feature engineering and modeling

Feature engineering creates user, item, and interaction features with explicit definitions. Modeling trains baseline, collaborative, and content-oriented candidates and evaluates held-out observations using recommendation-appropriate measures.

## 5. Implementation Approach

The implementation favors small, typed Python services and YAML-based behavioral configuration. Public boundaries validate inputs, persisted artifacts use stable locations, and tests isolate business behavior from network or infrastructure dependencies.

- Python 3.12 is the target runtime.
- Pandas and PyArrow support preparation and tabular artifacts.
- FastAPI exposes the popularity source contract.
- Scikit-learn supports preprocessing, baselines, and evaluation.
- ReportLab renders this configuration-driven academic PDF.

## 6. Reproducibility and Quality Assurance

Reproducibility is supported by version-controlled code, constrained dependencies, external configuration, deterministic random seeds, chronological split definitions, and explicit artifact schemas. Automated tests cover focused units and selected stage integrations.

Data quality is a pipeline gate rather than an informal inspection. Invalid records are identified by named rules, accepted and rejected counts are reconciled, and optional enrichment is distinguished from data required for core processing.

## 7. Assumptions and Limitations

MovieLens ratings are treated as product-preference observations, while generated commercial events are simulations rather than measurements from a production marketplace. Offline results therefore do not prove real-world business impact.

- The dataset is small and reflects an older film catalogue.
- Simulated clicks, purchases, prices, and trends inherit generator assumptions.
- Offline evaluation cannot directly measure satisfaction or long-term effects.
- Cold-start behavior is constrained without historical interactions.
- This version contains no dynamic execution evidence or pipeline metrics.

# 8. Future Work

Later versions can incorporate verified runtime evidence after the static submission structure is reviewed. Suitable extensions include model-comparison tables, quality summaries, lineage references, experiment identifiers, and sourced visualizations, while preserving the distinction between authored narrative and collected evidence.

# 9. Conclusion

RecoMart provides a foundation for studying recommendation systems as complete data products rather than isolated algorithms. Layered contracts, validation gates, reproducible configuration, and comparative modeling establish a clear path from educational source data to auditable recommendations. This report is maintainable and ready for later, evidence-based extension.

# References

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[3] ReportLab Software, ReportLab PDF Library User Guide.

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